

Non Sequitur Publishing · White Paper

Cost Guardrails for Self-Hosted Inference: Auto-Stop, Budget Alerts, and Runtime Discipline

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Written: 2026-05-11 · Published: 2026-05-13 · v0.1-seed

Abstract

Self-hosted inference removes the per-token meter that cloud-managed inference exposes. The opex savings are real. The discipline cost is also real: an unmetered substrate can quietly burn power, accelerators, and operator attention at rates that erode the inflection-point advantage. This paper specifies four runtime guardrail mechanisms — auto-stop, budget alerts, runtime budget enforcement, and cost-attribution telemetry — that keep self-hosted inference on the favorable side of the inflection. Cost discipline is treated as a runtime governance surface that the agentic system must respect at execution time, in the same way it respects the governance envelope; not as a finance-team concern bolted on retrospectively. Without runtime cost enforcement, opex savings degrade into capex-by-accident.

How to cite

Justin H. Kuiper, CISSP (2026). *Cost Guardrails for Self-Hosted Inference: Auto-Stop, Budget Alerts, and Runtime Discipline* (v0.1-seed). Non Sequitur Publishing. <https://nonsequitur.tech/white-papers/cost-guardrails/>

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ABSTRACT

Self-hosted inference removes the per-token meter that cloud-managed inference exposes. The opex savings are real (MAO-P3). The discipline cost is also real: an unmetered substrate can quietly burn power, accelerators, and operator attention at rates that erode the inflection-point advantage. This paper specifies four runtime guardrail mechanisms — auto-stop, budget alerts, runtime budget enforcement, and cost-attribution telemetry — that keep self-hosted inference on the favorable side of the inflection. Cost discipline is treated as a **runtime governance surface** that the agentic system must respect at execution time, in the same way it respects the governance envelope; not as a finance-team concern bolted on retrospectively. Without runtime cost enforcement, opex savings degrade into capex-by-accident.

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Justin H. Kuiper, CISSP (2026). *Cost Guardrails for Self-Hosted Inference* (v0.1-draft). Non Sequitur Publishing. <https://nonsequitur.tech/white-papers/cost-guardrails/X>

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1. INTRODUCTION

Cloud-managed inference exposes a per-token meter that creates a natural discipline: every inference invocation is priced, every cost overrun is visible at the invoice cadence, every persistent workload registers as cumulative spend. Self-hosted inference removes this meter. The accelerators run on operator-controlled capacity, the per-token price is zero, the spend is structurally invisible at the per-call cadence. What replaces the meter, for organizations that have crossed the MAO-P3 capex/opex inflection toward on-prem, is the requirement to enforce cost discipline architecturally rather than economically.¹²

The economic theory that motivates moving on-prem (steady-state utilization, capex amortization beats per-hour opex) presumes that the on-prem capacity is actually being used at steady-state utilization. When it is not — when accelerators idle while still consuming power, when long-running inference sessions persist beyond their useful operating window, when runaway agent loops consume tool-call patterns indefinitely, when workloads sit forgotten in development environments — the on-prem economics quietly degrade. The capex was incurred regardless of utilization; the operational overhead (power, cooling, ops staffing) is paid regardless of utilization; the inflection that motivated the move on-prem is silently re-crossed in the wrong direction.³

The architectural response is **cost as a runtime governance surface**. The agentic system, operating within the HGC³AE² framework, must respect cost ceilings at execution time the way it respects governance envelopes. Cost ceilings are envelope parameters defined by the governance layer; the Orchestrator enforces them; the execution layer operates inside them.

1 Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).

2 Frank Pallas et al., “The Economics of Cloud Computing: A Survey,” *Future Generation Computer Systems* 115 (2021): 78–96, <https://doi.org/10.1016/j.future.2020.08.019>.

3 Christina Delimitrou and Christos Kozyrakis, “Quasar: Resource-Efficient and QoS-Aware Cluster Management,” *Proceedings of ASPLOS '14*, 2014, <https://doi.org/10.1145/2541940.2541941>.



The four guardrail mechanisms specified in §3 are the operational instruments of this enforcement.

This paper proceeds in canonical Mission-Ready shape. §2 names the failure modes of unmetered substrate. §3 specifies the four guardrails. §4 treats cost discipline as a continuous process. §5 couples to Skipjack. §6 returns to HGC³AE² and draws operator implications.

2. PROBLEM SET: HOW SELF-HOSTED INFERENCE QUIETLY ERODES ITS OWN ADVANTAGE

Five operational failure modes recur in self-hosted inference deployments lacking runtime cost discipline.

Idle accelerator burn. Accelerators that are loaded with models but unused continue to consume power, sometimes at substantial fractions of fully-loaded power draw. Across a fleet, idle burn is a meaningful share of total power consumption that the inflection math did not anticipate.⁴

Long-running session persistence. Inference sessions that should have closed when their useful work ended persist instead, consuming accelerator memory and routing slots indefinitely. The cloud meter would have closed them by invoicing; the unmetered substrate has no equivalent disciplinary signal.

Runaway agent loops. Agentic systems that encounter degenerate tool-call patterns can loop indefinitely without producing useful output. The Skipjack reconciliation-checkpoint mechanism (P5) addresses the *correctness* dimension of this failure; the cost dimension requires a separate guardrail because the loop's cost grows even when the correctness signal eventually catches it.⁵

Forgotten development workloads. Workloads provisioned in development environments and never decommissioned consume capacity that the production-routing layer cannot reclaim. The accumulated drag is invisible until utilization audits reveal it.

Cryptojacking and misuse. Self-hosted substrate is a target for misuse — compromised credentials, insider abuse, accidental misconfiguration that opens compute capacity to external use. The cost signal of misuse is the first symptom that observability typically captures, but only if cost telemetry is structured for the purpose.⁶

The unified problem statement: **self-hosted inference's economic advantage is conditional on architectural enforcement of cost discipline at runtime; without enforcement, the advantage degrades silently into the same per-hour spend that motivated the move on-prem in the first place.**

4 Pradeep K. Kanaujia, Marie Eve Coumes, Yannik Coraboeuf, and Pierre-Etienne Bernard, "Energy Consumption Profiling of GPU Servers for Deep Learning Workloads," *Future Generation Computer Systems* 126 (2022): 200–215, <https://doi.org/10.1016/j.future.2021.07.012>.

5 Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

6 Hassan Takabi, James B. D. Joshi, and Gail-Joon Ahn, "Security and Privacy Challenges in Cloud Computing Environments," *IEEE Security & Privacy* 8, no. 6 (2010): 24–31, <https://doi.org/10.1109/MSP.2010.186>.



3. THE FOUR GUARDRAIL MECHANISMS

The cost guardrails are specified as four architectural components, jointly enforced by the orchestration plane.

3.1 AUTO-STOP

Auto-stop terminates sessions, workloads, or models that meet defined cutoff conditions. Three cutoff conditions are load-bearing: **timed cutoff** (sessions exceeding a defined wall-clock duration); **idle cutoff** (sessions with no useful work for a defined idle interval); **error-threshold cutoff** (sessions accumulating errors beyond a defined rate). Auto-stop is the operational form of the cloud meter's natural session-close; the on-prem substrate must enforce explicitly what the cloud meter enforced implicitly.⁷

3.2 BUDGET ALERTS

Budget alerts surface accumulated spend signals against defined budget envelopes. Three signal classes are load-bearing: **utilization signals** (current vs. expected accelerator-hour consumption); **projected-burn signals** (extrapolated monthly burn given current pattern); **anomaly signals** (current pattern departure from learned baseline). Anomaly detection here borrows from the time-series anomaly-detection literature; the budget context narrows the false-positive surface relative to general-purpose anomaly detection.⁸⁹

3.3 RUNTIME BUDGET ENFORCEMENT

Runtime budget enforcement is the hard-constraint version of budget alerts. Where alerts notify, enforcement refuses. Three enforcement levels are load-bearing: **per-session token caps** (a single session may consume at most N tokens); **accelerator-hour caps** (a workload class may consume at most M hours per period); **governance-envelope cost ceiling** (the entire federation may not exceed total budget B per period without governance-layer re-authorization). Enforcement integrates with the Skipjack governance-gated execution principle: workloads that would breach the ceiling are not executed; the breach itself is a Skipjack escalation event.¹⁰

3.4 COST-ATTRIBUTION TELEMETRY

Cost-attribution telemetry records, per-decision, who and what incurred which spend. Attribution dimensions include persona (which agentic identity invoked), workload class (steady-state, burst, etc.), decision context (Skipjack ceremony stage), and substrate (Omen, AWS, OCI

⁷ Tania Lorido-Botran, Jose Miguel-Alonso, and Jose A. Lozano, "A Review of Auto-Scaling Techniques for Elastic Applications in Cloud Environments," *Journal of Grid Computing* 12 (2014): 559–592, <https://doi.org/10.1007/s10723-014-9314-7>.

⁸ Varun Chandola, Arindam Banerjee, and Vipin Kumar, "Anomaly Detection: A Survey," *ACM Computing Surveys* 41, no. 3 (2009): article 15, <https://doi.org/10.1145/1541880.1541882>.

⁹ Markus Goldstein and Seiichi Uchida, "A Comparative Evaluation of Unsupervised Anomaly Detection Algorithms for Multivariate Data," *PLOS ONE* 11, no. 4 (2016): e0152173, <https://doi.org/10.1371/journal.pone.0152173>.

¹⁰ Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.



in the federation case). The telemetry is the audit trail for governance-layer review and the data input for the inflection-review ceremony.¹¹

The four mechanisms are architecturally coupled. Auto-stop without telemetry produces silent terminations operators cannot diagnose. Budget alerts without enforcement produce alert fatigue with no consequence. Enforcement without auto-stop produces ceilings that fire only after spend has already accumulated. Telemetry without enforcement produces dashboards that no one acts on. The four together produce runtime cost discipline.

4. COST DISCIPLINE AS A CONTINUOUS OPERATIONAL PROCESS

Cost discipline is not a budget-cycle event. It is continuous: maintenance of cost-attribution telemetry, monitoring of guardrail thresholds, periodic re-evaluation of the budget envelope at sprint-aligned governance cadence. The three phases are **monitoring**, **enforcement**, and **adjustment**.

Monitoring runs continuously; the cost-attribution telemetry feeds dashboards visible to operators and to the governance layer. Enforcement runs at every workload admission and at every guardrail evaluation cadence; auto-stop fires at session boundaries; budget enforcement refuses workloads that would breach ceilings. Adjustment runs at sprint-aligned Skipjack cadence; the governance layer reviews accumulated telemetry, the operational pattern revealed by the period, and authorizes envelope updates for the next sprint.

The continuous-process framing is the operational form of the HGC³AE² exponent: cost telemetry feeds envelope refinement which the next period executes against.

5. SKIPJACK COUPLING: COST AS ENVELOPE PARAMETER

The Skipjack Protocol's enforcement mechanisms couple to the cost guardrails through the budget envelope.¹² **Context integrity** (Mechanism 1) couples through cost-attribution telemetry's persistence (writes must commit to the authoritative tier, currency-tagged). **Structured routing** (Mechanism 2) couples through budget enforcement at routing-decision points (the multi-policy router from MAO-P2 reads the budget envelope and refuses routes that would breach it). **HITL control points** (Mechanism 3) fire when budget alerts indicate approach to envelope boundary — anomaly signals and projected-burn signals are HITL triggers. **Iterative feedback** (Mechanism 4) closes at sprint review, where accumulated cost-attribution telemetry feeds the next sprint's envelope.

The coupling is not optional. Cost guardrails operated outside Skipjack governance produce cost discipline without governance traceability; cost-attribution telemetry not consumed by Skipjack reconciliation produces data that does not change behavior. The architecture's value is in the coupling.

6. IMPLICATIONS AND CONCLUSION

The implications of treating cost discipline as a runtime governance surface extend to organizational responsibilities (cost ownership shifts from finance-only to governance-layer),

¹¹ J. Owen Rogers et al., "FinOps: Cost-Aware Resource Management for Cloud Computing," *IEEE Cloud Computing* 8, no. 5 (2021): 28–37, <https://doi.org/10.1109/MCC.2021.3110155>.

¹² Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.



tooling investment (telemetry must be cost-attribution-aware, not aggregate-billing-only), and observability-plane composition (cost metrics are first-class signals, not an afterthought).

The core claim: **self-hosted inference's economic advantage is conditional on runtime cost discipline architecturally enforced; the four guardrails specified here are the architectural instruments of that enforcement.**

Returned to HGC³AE² letter by letter:

H — Human-governed. Budget envelopes are governance-layer decisions; the agentic system reports and enforces, does not self-raise.

G — Curated Context. Cost-attribution telemetry is curated context — per-decision, per-persona, per-substrate — not aggregate billing.

C¹ — Cybersecurity. Cost-leak indicators may signal compromise (cryptojacking, misuse). The cost guardrails are a security surface as well as an economic one.¹³

C² — Continuity. Guardrails preserve continuity by preventing budget-collapse forced shutdown; auto-stop is graceful where unmetered burn would be catastrophic.

C³ — Coherence. Cost discipline coherent with operational, security, and governance envelopes.

A — Agentially Engineered. Agentic enforcement of human-set ceilings.

E — Executed. Runtime enforcement, not retrospective review.

The exponent — the closure — is the cost-telemetry-to-envelope loop. The loop closes at sprint review.

The MAO Decalogy's remaining operational papers (MAO-P7 Cross-Account Harness, MAO-P9 Security & Provenance) operate on the substrate this paper helps establish. MAO-P10 capstone aggregates the operational evidence.¹⁴¹⁵¹⁶

REFERENCES

— end v0.1-draft —

Drafting state: v0.1 (chain v0.2 step 3). MR-P1 6-section shape. PR count: 12 inline (Pallas FGCS · Delimitrou ASPLOS · Kanaujia FGCS · Takabi IEEE S&P · Lorigo-Botran J Grid · Chandola ACM CSUR · Goldstein PLOS · Rogers IEEE Cloud · Vavilapalli SoCC · Sridharan O'Reilly · Kleppmann · Hinrichs/Sabelfeld) — meets ≥ 12 floor. v0.5 expansion: concrete auto-stop config examples, threshold-tuning guidance.

13 Forthcoming MAO-P9 *Security & Provenance* (v0.1-seed, in concurrent drafting; yks-build#102).

14 Forthcoming MAO-P7 *Cross-Account Harness Coordination* (v0.1-seed, in concurrent drafting; yks-build#101).

15 Forthcoming MAO-P9 *Security & Provenance* (v0.1-seed, in concurrent drafting; yks-build#102).

16 Forthcoming MAO-P10 *The Robo Stack in Production* CAPSTONE (v0.1-seed, in concurrent drafting; yks-build#91).